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Atlantic Technological University

Tablet Classification App

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Project Engineering · Final Year Project 2025/26

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Project Overview

The Tablet Classification App is a mobile app used for identifying tablets. This project uses machine learning to help the user identify a tablet that they do not recognise. I have completed this by training the machine learning model in Google Colab on a dataset of 4000 tablet images and building a mobile app for the user to use this. My aim is to help people with memory problems to take control over this aspect of their lives.

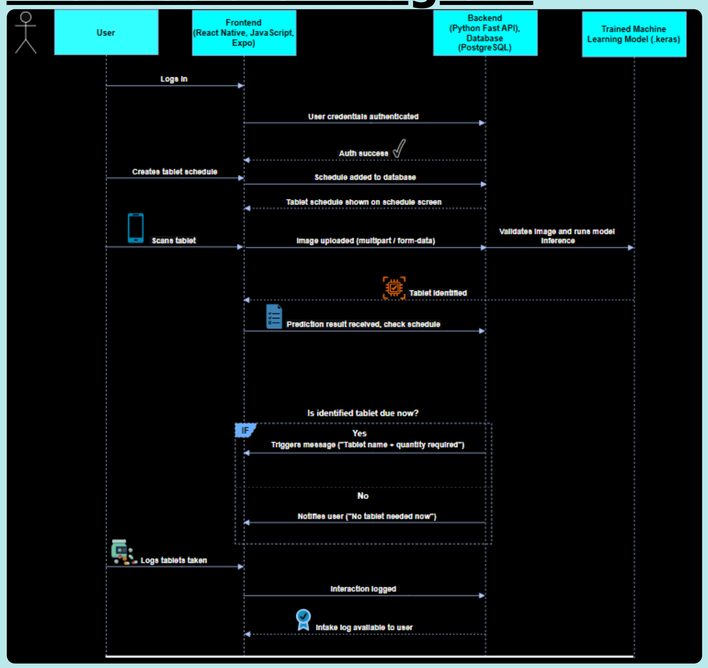
How it works:

1. User signs up/logs in to an account.
2. The user can navigate to the schedules screen where they create a tablet schedule which includes tablet names, time it's due and the quantity required.
3. The tablet prediction screen allows the user to open the phone camera or upload an image of a tablet to the prediction endpoint to identify the image.
4. When the tablet is identified the user is told if it is required at this time, according to their schedule.
5. If the tablet is taken, the user can log this activity with one press of a button to a PostgreSQL database, so the user can track their own intake status.

Machine Learning

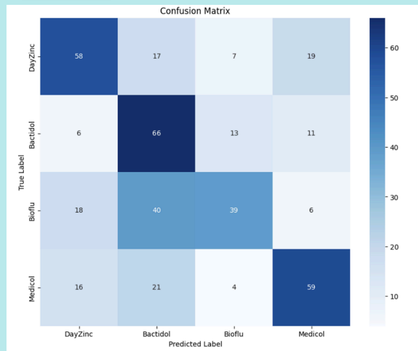
Machine learning model was trained in Google Colab using a dataset which I imported from Kaggle of 4000 tablet images. I have loaded in a pretrained ResNet50 model to achieve high accuracy for classification and detection, rapidly. Using a pretrained model saves significant training time as the model already understands basic visual features like edges and shapes.

Architectural Diagram

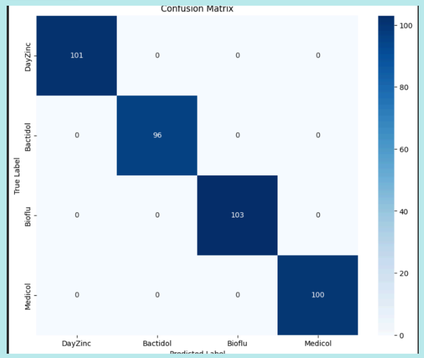


Results

Before:

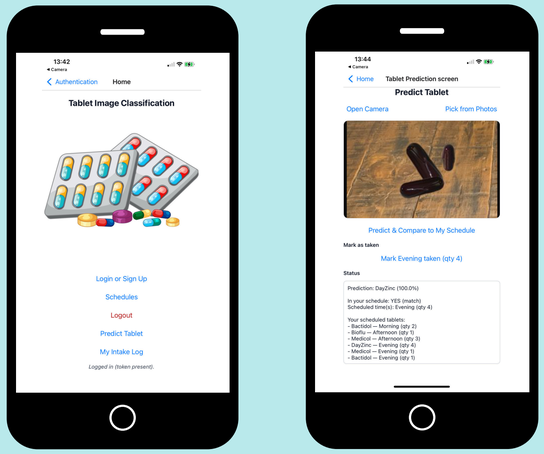


After:



The Y-axis of the confusion matrix shows what the tablet actually is, the X-axis shows what the model has predicted the tablet as. The confusion matrix is plot based on the results of the 400 images used for testing The image on the right is the desirable end goal which I achieved after many weeks of bug-fixing and hyperparameter tuning.

Application in use



The image on the left is the home screen that the user will see once logged in, giving them options to view their tablet schedule, intake log, or to scan a tablet to identify it. The image on the right is the prediction screen, where the user can open the camera or upload an image to have it identified

Software used

- FastAPI: Python web framework used to build APIs in Visual Studio Code.
- React Native: Frontend framework for building user interfaces.
- JavaScript: Frontend programming language.
- Expo: A full-stack React Native framework.
- PostgreSQL: Database.
- Keras: Deep learning API used to build, train and deploy neural networks in Google Colab.

Conclusion

The Tablet Classification App successfully demonstrates that machine learning can be used to help people with memory difficulties manage their medication safely. The core goal of reducing human error in tablet-taking has been achieved, and the project demonstrates a practical, real-world application of machine learning in healthcare.